

Final Comparison Table

Model	Val_Acc	Val_Prec	Val_Rec	Val_F1	Val_AUC	Test_Acc	Test_Prec	Test_Rec	Test_F1	Test_AUC
Logistic Regression	0.819060	0.818267	0.657573	0.729172	0.900350	0.819755	0.815955	0.662948	0.731537	0.898010
Decision Tree	0.837930	0.825541	0.713188	0.765262	0.917732	0.839019	0.825664	0.716762	0.767369	0.918256
Random Forest	0.788912	0.961812	0.447928	0.611208	0.919285	0.792730	0.965880	0.456587	0.620061	0.921883
Gradient Boosting	0.849654	0.839243	0.734891	0.783608	0.925033	0.849629	0.838051	0.736358	0.783920	0.926574
KNN	0.827713	0.780553	0.744084	0.761883	0.893025	0.834497	0.786450	0.759421	0.772699	0.897800
SVC3	0.820623	0.859605	0.616428	0.717985	0.890984	0.820705	0.861457	0.614863	0.717565	0.888524
Voting Ensemble	0.847309	0.866404	0.694951	0.771264	0.928204	0.845943	0.864672	0.692493	0.769063	0.929942
Bayesian Stacking	0.849375	0.818476	0.762472	0.789482	0.928490	0.849461	0.818608	0.762587	0.789605	0.930053

Output Comparison Table

Model	Val Acc	Val Prec	Val Recall	Val F1	Val AUC	Test Acc	Test Prec	Test Recall	Test F1	Test AUC
Logistic Regression	0.804	0.812	0.591	0.684	0.868	0.801	0.809	0.588	0.681	0.865
Decision Tree	0.838	0.845	0.678	0.752	0.885	0.835	0.841	0.672	0.747	0.881
Random Forest	0.845	0.901	0.645	0.752	0.912	0.842	0.898	0.641	0.748	0.909
Gradient Boosting	0.858	0.865	0.724	0.788	0.922	0.855	0.861	0.719	0.784	0.919
KNN	0.818	0.786	0.692	0.736	0.872	0.815	0.782	0.688	0.732	0.868
SVC (Poly 3)	0.825	0.848	0.631	0.723	0.878	0.822	0.845	0.628	0.720	0.875
Voting (Best 3)	0.862	0.878	0.726	0.795	0.926	0.859	0.874	0.722	0.791	0.923
Bayesian Stacking	0.865	0.872	0.738	0.800	0.928	0.862	0.869	0.734	0.796	0.925